

Long-Tail Rebalancing for Non-Verbal Vocalization-Aware ASR: A Track 1 System for the NVVSpeech Challenge

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Abstract

Non-verbal vocalizations (NVVs) carry important paralinguistic information but are often omitted by conventional automatic speech recognition (ASR) systems. The ISCSLP NVVSpeech Challenge requires joint transcription of lexical content and 16 NVV categories under limited and highly imbalanced supervision. We present a data-centric NVV-aware ASR pipeline based on cross-dataset label harmonization and a two-stage sampling schedule. We map heterogeneous source labels to the official taxonomy and exclude samples without a reliable mapping. Our schedule first uses square-root category sampling to moderate the long-tailed distribution and then applies uniform-category fine-tuning. On a fixed local validation split, square-root category sampling performs best among the tested single-stage settings. The final two-stage system obtains an official score of 63.86 and ranks fourth in Track 1.

Index Terms: speech recognition, non-verbal vocalization recognition, computational paralinguistics

1. Introduction

Non-verbal vocalizations (NVVs), such as laughter, coughing, breathing, and crying, convey information about speaker state, affect, and conversational dynamics beyond lexical content [1, 2, 3, 4, 5]. Conventional automatic speech recognition (ASR) systems, however, primarily focus on lexical transcription and often omit these events. Consequently, lexical-only transcripts fail to capture how speakers feel, react, and interact.

Recent NVV-annotated datasets have expanded the coverage of non-verbal speech events through scripted collection, human annotation, and automatic labeling of real-world speech [6, 7, 8, 9, 10]. However, these datasets differ in event inventories, annotation granularity, and data provenance, making joint training across datasets nontrivial. In parallel, event-aware ASR systems have begun to represent NVV events as inline output tokens, enabling lexical content and non-verbal events to be recognized within a unified decoding sequence [11, 12, 13, 14]. The ISCSLP NVVSpeech Challenge provides a concrete testbed for this problem by requiring systems to jointly transcribe speech and 16 predefined NVV categories [15]. Nevertheless, normalizing heterogeneous labels alone does not resolve the severely long-tailed category distribution: simply combining all available data leaves rare NVV

categories underrepresented. At the other extreme, aggressively flattening the category distribution does not necessarily improve the aggregate score. This sensitivity motivates the question of how heterogeneous NVV resources should be normalized and rebalanced without distorting the joint lexical–NVV prediction task.

To address this problem, we collect several open-source datasets, map their NVV labels to the 16 challenge categories, and exclude samples without a reliable mapping. We then fine-tune Qwen3-ASR with a two-stage schedule. Stage 1 uses square-root category sampling to smooth the global distribution, and Stage 2 continues training with uniform-category sampling to place greater emphasis on tail categories.

We evaluate sampling strategies on a fixed local validation split using the official Track 1 evaluator. Square-root sampling gives the highest local validation score among the tested single-stage settings, whereas stronger rebalancing is not monotonically better. The two-stage system gives our highest official submission score, but the single-stage square-root model scores higher on the local validation split.

Our contributions are threefold:

- We harmonize heterogeneous NVV annotations into the 16-category challenge taxonomy and filter unreliable mappings for joint lexical–NVV modeling.
- We systematically study category-level rebalancing strength under a fixed backbone, objective, and output format, showing that moderate square-root sampling outperforms both natural and fully uniform sampling on our local validation split.
- We provide complementary evaluation across local validation, the official challenge test, and a held-out-speaker MNV-17 test set, and our final two-stage system achieves an official score of 63.86, ranking fourth in Track 1.

2. Related work

2.1. Event-aware speech recognition

Most ASR systems prioritize lexical transcription, whereas event-aware systems also include social or non-verbal events in the output sequence. End-to-end studies have jointly modeled speech recognition and social-signal detection and later integrated lexical and non-verbal information in unified recognition frameworks [11, 12]. More recent work combines NVV data construction, recognition, and generation by treating NVVs as decodable elements [6, 7, 16]. WESR-Bench further defines

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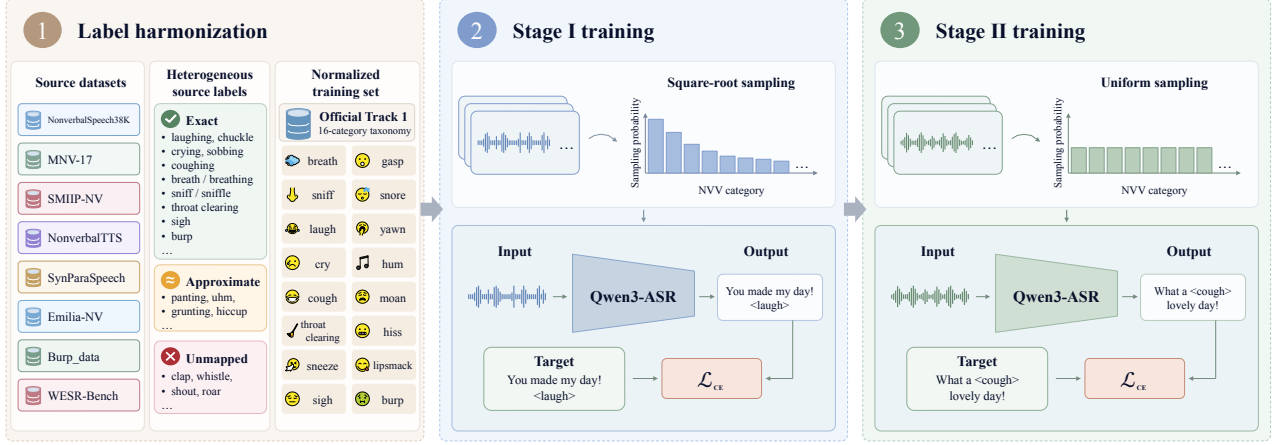


Figure 1: Overview of our data-centric NVV-aware ASR pipeline, consisting of cross-dataset label harmonization, Stage I training with square-root category sampling, and Stage II fine-tuning with uniform-category sampling.

explicit event categories and position-aware evaluation [13]. In contrast, we keep the backbone and output protocol fixed and study training-data composition and category sampling.

Data-centric strategies such as two-stage training and class balancing can improve rare-event recognition [17]. However, the setting in which multiple heterogeneous datasets must first be mapped into a challenge taxonomy remains less explored. We address this setting without an additional event detector or augmentation model.

2.2. Non-verbal vocalization datasets and label harmonization

NVV datasets differ in event inventories, annotation granularity, languages, and collection procedures. NonverbalSpeech38K provides large-scale data for non-verbal speech understanding and generation [7]; MNV-17 focuses on performative Mandarin events [9]; and SMIIP-NV provides multiply annotated expressive Mandarin speech [8]. The challenge and WESR-Bench highlight the need for a unified label space and a consistent evaluation protocol [13, 15]. Direct merging can conflate related but non-equivalent labels and amplify imbalance because the datasets differ in scale and category coverage. We therefore map direct and manually verified correspondences to the Track 1 taxonomy and exclude labels without a reliable mapping.

2.3. Class imbalance and data rebalancing

Long-tailed recognition work has compared instance-balanced, class-balanced, square-root, and progressively balanced sampling. Square-root sampling provides an intermediate distribution, whereas progressive balancing changes the distribution during training [18]. Dynamic Curriculum Learning jointly adjusts sampling and loss schedules [19], and Balanced Meta-Softmax estimates class sampling rates from meta-validation data [20]. We instead keep the objective fixed and use a simple two-stage schedule: square-root category sampling followed by uniform-category fine-tuning.

3. Method

3.1. Problem formulation

We formulate NVV-aware ASR as conditional sequence generation. For an audio waveform x and language identifier ℓ , the model produces one sequence y that interleaves lexical words with inline NVV tags. Let $\mathcal{D} = \{(x_i, \ell_i, y_i)\}_{i=1}^N$ denote the processed training set, and let q denote the example distribution induced by the sampler. The ASR model is trained with the standard autoregressive supervised fine-tuning (SFT) objective

$$\mathcal{L}_{\text{SFT}}(\theta; q) = -\mathbb{E}_{i \sim q} [\log p_{\theta}(y_i | x_i, \ell_i)], \quad (1)$$

while keeping the backbone, tokenizer, target serialization, and loss fixed. We modify only q through category-aware sampling. The method combines cross-dataset label harmonization, power-based category sampling, and a two-stage full-parameter SFT schedule.

3.2. Cross-dataset label harmonization

Merging NVV datasets directly is problematic because the same event may appear under different names, while one source label may be broader or less reliable than an official Track 1 category. At the same time, dropping every source label that is not textually identical would discard useful supervision. We therefore use the official 16-category Track 1 label set as a common target space and define a fixed mapping from source labels into it, as illustrated in the label-harmonization stage of Figure 1.

Specifically, for a source label s , we map it to the corresponding official category when the correspondence is exact or a manually verified approximation; labels without a reliable interpretation are left unmapped and excluded from the harmonized training set. For utterances containing multiple events, we normalize each event label independently while retaining the utterance as a single training example, thereby preserving the complete lexical-NVV sequence and the original event order. The resulting harmonized training set is summarized in Table 1.

3.3. Power-based category sampling

As shown in Figure 2, the category distribution remains long-tailed even after harmonization. Natural sampling, which fol-

Table 1: *Composition of the normalized NVV-aware ASR training data.*

Dataset	Lang.	Samples	Hours
NonverbalSpeech38K [7]	ZH / EN	9,597	31.88
MNV-17 [9]	ZH	1,497	4.26
Burp_data ¹ [15]	ZH / EN	1,425	2.11
NonverbalTTS [21]	EN	3,518	9.14
SMIP-NV [8]	ZH	3,008	6.06
SynParaSpeech [22]	ZH	4,000	5.51
Emilia-NV [23]	ZH	3,011	10.02
WESR-Bench [13]	EN	592	1.50
Total	ZH / EN	26,648	70.48

lows the empirical frequencies of the assigned categories ($\alpha = 1$), gives the model many updates on frequent events but few opportunities to learn the acoustic cues and corresponding NVV labels of rare events. Inspired by class-level rebalancing practice [18], we address this mismatch using category-level sampling while leaving each utterance intact.

Let c_i denote the sampling category assigned to utterance i , and let n_c denote the number of utterances assigned to category c . For utterances containing multiple valid NVV events, we use the first valid NVV label in target order as the sampling key, while keeping the complete lexical–NVV target sequence unchanged. Only 1,413 of 26,648 training utterances (5.30%) contain multiple distinct valid NVV labels, suggesting that this assignment has a limited effect on the overall rebalancing distribution. Let $\mathcal{C}_{\text{train}} = \{c \mid n_c > 0\}$ denote the represented categories. For $\alpha \in [0, 1]$, we define

$$p_c(\alpha) = \frac{n_c^\alpha}{\sum_{k \in \mathcal{C}_{\text{train}}} n_k^\alpha}, \quad (2)$$

At each sampling step, we first sample $c \sim p_c(\alpha)$ and then uniformly sample an utterance from the corresponding category bucket. Hence, the induced utterance-level sampling probability is

$$q_\alpha(i) = \frac{p_{c_i}(\alpha)}{n_{c_i}}. \quad (3)$$

When $\alpha = 1$, q_α reduces to uniform sampling over training utterances; $\alpha = 0.5$ gives square-root sampling (SQRT) [24], while $\alpha = 0$ assigns equal probability to each represented category.

3.4. Two-stage sampling schedule

As illustrated in Figure 1, our final schedule consists of square-root sampling in Stage 1 followed by uniform-category sampling in Stage 2. In **Stage 1**, we fine-tune Qwen3-ASR 1.7B with full-parameter updates while moderately increasing the exposure of tail categories. In **Stage 2**, we resume from the Stage 1 checkpoint and continue SFT with equal category probabilities. The backbone, objective, target serialization, and lexical–NVV sequence format remain unchanged across stages.

4. Experiments

4.1. Experimental setup

Backbone selection. Qwen3-ASR provides a multilingual ASR backbone with strong Chinese and English recognition capabilities [25], making it suitable for the challenge test set, which

¹https://huggingface.co/datasets/NVVSpeech-Challenge/Burp_data

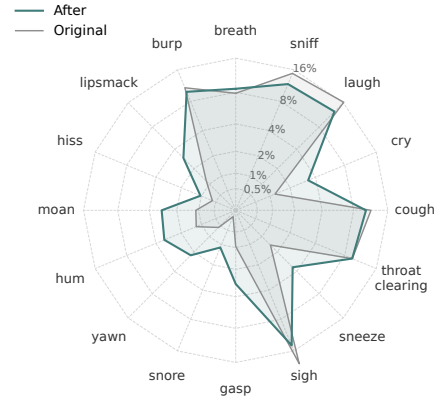


Figure 2: *NVV-category occurrence shares in the original training set and after square-root resampling. The radial axis uses a logarithmic transformation to better visualize rare-category exposure.*

contains approximately balanced numbers of Chinese and English utterances. Of the two released variants, 0.6B and 1.7B, we use the 1.7B model for full-parameter SFT.

Training and inference. We train with bfloat16 precision, a global batch size of 64, a learning rate of 2×10^{-5} with linear decay, a warmup ratio of 0.02, and a gradient clipping threshold of 1.0. In our main two-stage system, Stage 1 and Stage 2 each run for 5 epochs. For inference, we use the released Qwen3-ASR implementation with greedy decoding. Local validation outputs are scored using the official Track 1 evaluator.

Evaluation data. We evaluate our systems on a local validation split, the official challenge test set through submission, and a subset of the public MNV-17 test split. For local model selection, we use a fixed validation split of 214 utterances: 147 Chinese and 67 English utterances covering all 16 categories. These comprise 108, 86, and 20 utterances from NonverbalSpeech38K, MNV-17, and Burp_data, respectively. The MNV-17 test subset is used to assess generalization to unseen speakers, as detailed in Section 4.4.

Evaluation metrics. Following the Track 1 protocol [15], we use character error rate (CER) and word error rate (WER) to measure Chinese and English lexical transcription errors, respectively. We also report the NVV recognition F_1 score, multi-event normalized tag distance (mNTD), and tagged-transcript error (Err.). On MNV-17, we follow the original benchmark [9] and report joint CER and NVV accuracy, with the latter requiring an exact match between the complete predicted and reference NVV sequences.

4.2. Main results and analysis

Table 2 and Table 3 summarize the results on our local validation set and the official challenge test set, respectively. The Stage 1 SQRT system achieves the best local validation score of 66.25 and obtains an official score of 52.61 in the Preliminary Stage. In contrast, the Stage 2 model achieves a lower local validation score of 58.91, but obtains our best official result of 63.86 in the Final Stage.

Table 4 traces how per-category recognition changes from the SQRT checkpoint to the two-stage checkpoint. Yawn shows the clearest improvement, with F_1 moving from 0.09 to 0.39,

Table 2: Main results on the local validation split using the official Track 1 evaluator. We compare the released Whisper baseline with Qwen3-ASR 1.7B trained using natural sampling, square-root sampling (SQRT), and the two-stage SQRT + uniform schedule. The 10-epoch SQRT setting uses the same total number of training epochs as the two-stage system. We report the overall validation score, along with separate scores and evaluation metrics for Chinese and English.

System	Setting	Validation score $\uparrow$	Chinese					English				
			Score $\uparrow$	F ₁ $\uparrow$	mNTD $\downarrow$	Err. $\downarrow$	CER $\downarrow$	Score $\uparrow$	F ₁ $\uparrow$	mNTD $\downarrow$	Err. $\downarrow$	WER $\downarrow$
Official baseline	Released Whisper	37.26	48.68	0.43	0.49	0.19	13.67%	25.84	0.20	0.81	0.21	11.37%
Qwen3-ASR 1.7B	Natural ($\alpha = 1$), 5 epochs	62.33	67.83	0.66	0.38	0.06	3.60%	56.83	0.53	0.47	0.11	6.31%
Qwen3-ASR 1.7B	SQRT ($\alpha = 0.5$), 5 epochs	66.25	71.06	0.69	0.32	0.06	3.39%	61.45	0.59	0.43	0.11	7.10%
Qwen3-ASR 1.7B	SQRT + uniform, 5 + 5 epochs	58.91	65.52	0.62	0.35	0.06	3.79%	52.29	0.48	0.52	0.11	6.95%
Qwen3-ASR 1.7B	SQRT ($\alpha = 0.5$), 10 epochs	62.72	68.31	0.66	0.36	0.06	3.57%	57.12	0.54	0.47	0.11	6.16%

Table 3: Official Track 1 baseline and submission scores.

System	Challenge phase	Official score
Official Whisper baseline	Final	33.32
Qwen3-ASR 1.7B, SQRT	Preliminary	52.61
Qwen3-ASR 1.7B, two-stage	Final	63.86

while throat clearing moves from 0.50 to 0.57. SQRT remains stronger for most other categories, so the two-stage system changes the balance of recognition rather than lifting performance uniformly across the label set.

Table 4: Per-category local-validation results for the SQRT and two-stage systems. Val gives the number of reference occurrences for each category; R denotes recall, and ΔF_1 denotes the change from SQRT to two-stage using the displayed F₁ values.

Category	Val	SQRT F ₁ /R	Two-stage F ₁ /R	ΔF_1
sigh	20	0.69/0.85	0.64/0.75	-0.05
laugh	20	0.90/0.95	0.82/0.80	-0.08
sniff	20	0.61/0.70	0.47/0.45	-0.14
cough	20	0.39/0.35	0.26/0.20	-0.13
burp	20	0.95/0.90	0.76/0.80	-0.19
throat clearing	20	0.50/0.75	0.57/0.80	+0.07
breath	20	0.63/0.65	0.32/0.25	-0.31
sneeze	10	0.89/0.80	0.86/0.90	-0.03
hum	11	0.96/1.00	0.79/1.00	-0.17
cry	20	0.44/0.40	0.37/0.25	-0.07
lipsmack	13	0.96/1.00	0.80/0.92	-0.16
moan	11	0.91/0.91	0.71/1.00	-0.20
gasp	20	0.30/0.20	0.19/0.15	-0.11
hiss	7	0.83/0.71	0.56/1.00	-0.27
yawn	20	0.09/0.05	0.39/0.30	+0.30
snore	15	0.81/0.73	0.76/0.73	-0.05

4.3. Ablation of rebalancing strength

To investigate how the strength of category rebalancing affects NVV-aware ASR performance on the local validation set, we compare the validation scores under different sampling exponents α , ranging from uniform category sampling ($\alpha = 0$) to natural sampling ($\alpha = 1$). As shown in Figure 3, the best score of 66.25 is achieved at $\alpha = 0.5$, whereas $\alpha = 0.75$ gives the lowest score of 60.69, suggesting that moderate rebalancing is more effective than either weaker or stronger rebalancing on our local validation set. As shown in Table 2, compared with natural sampling, square-root sampling improves NVV recognition, while the English WER increases from 6.31% to 7.10%, indicating a modest trade-off between NVV recognition and lexical transcription performance. Meanwhile, uniform sampling ($\alpha = 0$) is suboptimal, suggesting that retaining some original

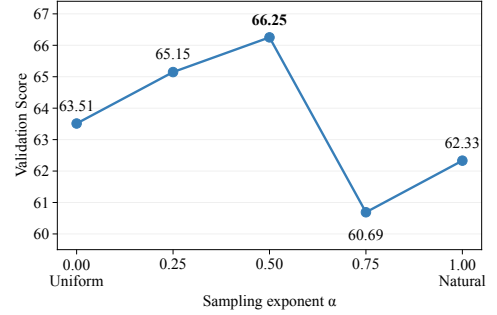


Figure 3: Local-validation score under different rebalancing strengths, indexed by α , under the same backbone and optimization recipe.

imbalance may be beneficial.

4.4. Evaluation on MNV-17

For generalization, we evaluate the baseline, SQRT, and two-stage systems on 140 Track-1-compatible Mandarin utterances from MNV-17. As shown in Table 5, the two-stage checkpoint achieves the lowest joint CER, while SQRT obtains higher NVV exact-match accuracy.

Table 5: Results on the MNV-17 test split.

System	Joint CER $\downarrow$	NVV accuracy $\uparrow$
Official Whisper baseline	21.49%	25.00%
Qwen3-ASR 1.7B (SQRT)	3.67%	63.57%
Qwen3-ASR 1.7B (two-stage)	3.21%	60.71%

5. Conclusion

Our data-centric NVV-aware ASR pipeline combines label harmonization, power-based sampling, and uniform-category fine-tuning without changing the backbone, objective, or output format. Square-root sampling performs best on the local validation split among the tested single-stage settings, while the two-stage system achieves our highest official submission score. On MNV-17, the two-stage model yields lower joint CER, whereas SQRT gives higher NVV exact-match accuracy. These results support category sampling as a practical data-centric tool for NVV-aware ASR. For future work, we plan to extend our approach to broader and more diverse datasets to further validate its generalizability and effectiveness.

6. References

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